ICMVA 2024

2024 The 7th International Conference on

MACHINE VISION AND APPLICATIONS

SINGAPORE

March 12-14, 2024

Organized by

SCIEI SCIENCE AND ENGINEERING INSTITUTE

Sensors and Systems Society of Singapore

Supported by

UNIVERSITY OF CANBERRA

岡山大学 OKAYAMA UNIVERSITY

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TAIWAN TECH NATIONAL TAIWAN UNIVERSITY OF SCIENCE AND TECHNOLOGY

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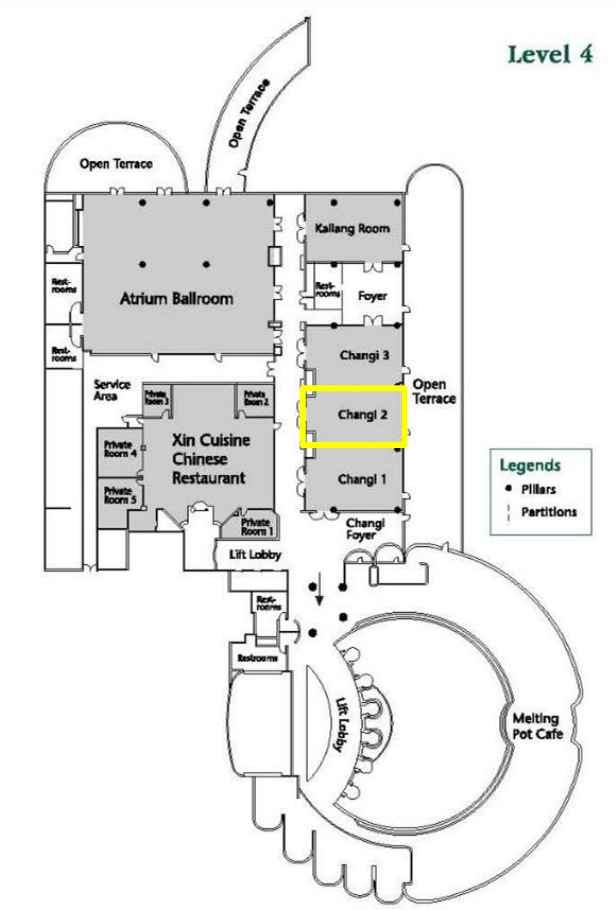
Conference Venue

Holiday Inn® Singapore Atrium

Address: 317 Outram Road Singapore 169075
Web: https://singaporeatrium.holidayinn.com/

An established brand under IHG, Holiday Inn Singapore Atrium is located in the heart of the city with Havelock MRT Station (subway) right outside its premise.
With Havelock MRT Station located just 3 minutes away, it provides direct access to Maxwell, Gardens by the Bay, Shenton Way, Orchard and Marina Bay. It is also just a short drive away to Sentosa and the Central Business District (CBD), and a 25 minute drive from Changi Airport.



Presentation Guideline

1 Onsite Registration

Registration desk (Foyer of Changi Room 2--4th floor)→ Inform the staff of your paper ID→ Sign-in→ Claim your conference kit.

2 Devices Provided by the Organizer

Laptops (with MS-Office & Adobe Reader) / Projectors & Screen / Laser Sticks

3 Materials Provided by the Presenter

Oral Session: Slides (pptx or pdf version).

Poster Session: Printed Poster (A1 Size).

Official Language: English.

4 Duration of Each Presentation

Keynote Speech: 40min, including Q&A.

Invited Speech: 30min, including Q&A.

Oral Session: 15min, including Q&A.

5 Notice

※ Please wear your delegate badge (name tag) for all the conference activities. Lending your participant card to others is not allowed.

※ Please take good care of your valuables at any time during the conference. The conference organizer does not assume any responsibility for the loss of personal belongings of the participants during conference day.

※ **UTC+8. Singapore Time. Please be aware of time difference between this and your region/country.**

6 Online Presentation Tips

	Meeting ID	Link
zoom	82976000510	https://us02web.zoom.us/j/82976000510

Note:

We recommend that you install the Zoom platform on your computer before the conference starts. New users can participate in the Zoom meeting without registration.

Participants who are going to do an online presentation are required to join the rehearsal in Zoom on **Mar. 12**. Feel free to leave after you finish the test.

◆Name Setting

Author: Session Number-Paper ID-Name

Listener: Listener-Name

Welcome Message

Welcome you all to The 7th International Conference on Machine Vision and Applications (ICMVA 2024), which will be held in Singapore during March 12-14, 2024. The conference is sponsored by Science and Engineering Institute (SCIEI), Sensors and Systems Society of Singapore (SSS), supported by Okayama University, University of Évora, Gifu Shotoku Gakuen University, and National Taiwan University of Science and Technology.

After several rounds of review procedure, the program committee accepted those papers to be published in conference proceedings. We wish to express our sincere appreciation to all the individuals who have contributed to ICMVA 2024 in various ways. Special thanks are extended to our colleagues in the program committees for their thorough review of all the submissions, which is vital to the success of the conference, and also to the members in the organizing committees who had devoted their time and efforts in planning, promoting, organizing and helping the conference.

One best presentation will be selected from each session, evaluated from: originality; applicability; technical Merit; qualities of PPT; English. The best one will be announced at the end of each session. Hope all of you can enjoy the conference.

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Organizing Committees

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Piyaporn Nurarak, Sukhothai Thammathirat Open University
Qiang Cheng, University of Kentucky
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Ranran Feng, University of Texas at Dallas
Robert Laramee, University of Nottingham
Sachin Kumar, Kyungpook National University
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Yap Hwa Jen, University of Malaya
Yasar Becerikli, Kocaeli University
Yongjun Zhang, Guizhou University
Zahid Akhtar, State University of New York Polytechnic Institute

Conference Schedule

Mar. 12, 2024 (Tuesday, UTC+8) | Changi Room 2 (4th floor)

10:00-12:00	**Online ZOOM Test (Authors & Presenters)**	**ZOOM ID: 82976000510**
10:00-12:00	**Onsite Registration & Materials Collecting**	**Foyer of Changi Room 2**
13:30-13:35	**Opening Remarks**	
13:35-13:40	**Welcome Address**	
13:40-14:20	**Keynote Speech I** *Prof. Yuan Luo* *National Taiwan University*	Abstract Page: 11
14:20-14:50	**Invited Speech I** *Mr. Max Liebmann,* *HOLOEYE Photonics AG*	Abstract Page: 13
14:50-15:20	**Invited Speech II** *Prof. Yoshio Hayasaki* *Utsunomiya University*	Abstract Page: 14
15:20-16:00	**Group Photo & Break**	**Foyer**
16:00-16:30	**Invited Speech III** *Prof. Chau-Jern Cheng* *National Taiwan Normal University*	Abstract Page: 15
16:30-17:00	**Invited Speech IV** *Prof. Wei Wang* *Heriot-Watt University*	Abstract Page: 16
17:00-17:30	**Invited Speech V** *Prof. Xiang Peng* *Shenzhen University*	Abstract Page: 17
17:30-18:00	**Invited Speech VI** *Prof. Rong Su* *Shanghai Institute of Optics and Fine Mechanics, Chinese Academy of Science*	Abstract Page: 18
18:00-18:30	**Invited Speech VII** *Prof. Kemao Qian* *Nanyang Technological University*	Abstract Page: 19
18:30-20:00	**Dinner**	**Atrium Restaurant**

Mar. 13, 2024 (Wednesday, UTC+8) | Changi Room 2 (4th floor)

9:00-9:40	**Keynote Speech II** *Prof. Yu Fu* *Shenzhen University*	Abstract Page: 12
9:40-10:10	**Invited Speech VIII** *Prof. Wenqi He* *Shenzhen University*	Abstract Page: 20
10:10-10:40	**Invited Speech IX** *Prof. Percival Almoro* *University of the Philippines Diliman*	Abstract Page: 21
10:40-11:00	**Group Photo & Break**	**Foyer**
11:00-11:30	**Invited Speech X** *Prof. Peng Gao* *Xidian Univeristy*	Abstract Page: 22
11:30-12:00	**Invited Speech XI** *Prof. Caojin Yuan* *Nanjing Normal University*	Abstract Page: 23
12:00-12:30	**Invited Speech XII** *Prof. Jun Ma* *Nanjing University of Science and Technology*	Abstract Page: 24
12:30-14:00	**Lunch**	**Atrium Restaurant**
14:00-16:00	**Session 1** Visual Based Object Recognition and Detection Technology	Page: 25
16:00-16:30	**Break & Poster Session**	**Foyer**
16:30-18:15	**Session 2** Image Based Data Analysis and Application System	Page: 26
18:30-20:00	**Dinner**	**Atrium Restaurant**

Mar. 14, 2024 (Thursday, UTC+8)

10:30-12:00	**Session 3 (Online Only)** Intelligent Image Analysis and Methods	Page: 27
12:00-14:00	**Lunch**	
14:00-15:30	**Session 4 (Online Only)** Image and Data Analysis Based on Machine Learning	Page: 28

Keynote Speaker I

Prof. Yuan Luo

National Taiwan University

Title: Advanced Optical Systems through Flat Optics for Biomedical Applications

Time: 13:40-14:20, Mar. 12 | **Onsite Room:** Changi Room 2 | **ZOOM ID:** 82976000510

Biography: Yuan Luo received the MSc and PhD degrees in the College of Optical Sciences in 2007 and 2008, respectively, from the University of Arizona. After post-doctoral work at the Massachusetts Institute of Technology (MIT, 2009–2011), he joined the faculty at National Taiwan University (NTU) in 2011, where he is now Director and Professor of the Institute of Medical Device and Imaging (MDI) at NTU College of Medicine. He also serves as Program Director of Precision Health and Intelligent Medicine (PHIM) at the NTU Graduate School of Advanced Technology and holds the Associate Deanship in the NTU YongLin Institute of Health. He has worked or held visiting appointments at the Singapore-MIT Alliance for Research and Technology (SMART) Center in Singapore. His research interests are in the fields of three-dimensional (3D) and spectral imaging, shift-invariant optics theory, nanophotonics, the design of diffractive optical elements, and the implementation of 3D active structured light. He is a Fellow of the International Society for Optical Engineering (SPIE).

Abstract: Optical imaging techniques are the most commonly used methods in biology and medical research. In recent years, structured light plays vital roles to enhance resolution. In addition, nano-photonics, the emerging optical techniques, with unique optical capabilities to manipulate the basic characteristics of light have received a significant amount of interest in the field of optical microscopy and endoscopy. This talk will introduce the latest studies on the biomedical use of structured light, as well as metasurfaces. The following are the topics that will be covered: quantitative cell imaging, optical microscopy; light sheet microscopy; and optical endoscopy. This talk will also discuss the technological challenges presently encountered with metasurface from the point of view of preclinical and clinical systems.

Keynote Speaker II

Prof. Yu Fu

Shenzhen University

Title: Structural Defects Localization using an LDV-enhanced Imaging Technology

Time: 9:00-9:40, Mar. 13| **Onsite Room:** Changi Room 2 | **ZOOM ID:** 82976000510

Biography: Dr. Yu Fu received his bachelor's in Mechanical Engineering from Shanghai Jiao Tong University and his Master's and Doctoral Degree in Mechanical Engineering from the National University of Singapore.

In 1997, Dr Fu joined the Department of Mechanical Engineering at the National University of Singapore as a Professional Officer, pursuing research on optical measurement and image processing techniques. He has published more than 100 publications in international leading journals and proceedings. In 2006, he was awarded the prestigious Alexander von Humboldt research fellowship and worked with the well-known applied optics research institute, Institut für Technische Optik (ITO), the University of Stuttgart, on optical dynamic measurement. He joined Temasek Laboratories of Nanyang Technological University in 2009. In July 2011, Dr. Fu received the prestigious Temasek Fellowship from Singapore's Defence Ministry and Nanyang Technological University to be a Principal Investigator in Temasek Laboratories. In 2018, he joined Shenzhen University as a Professor in the College of Physics and Optoelectronic Engineering. At present, Dr Fu is doing research in the area of optical dynamic measurement and structure health diagnostics. Dr Fu is also a Fellow member of SPIE.

Abstract: Vibration measurement, particularly mode shape measurement, is important to structural dynamic analysis and defect localization since it can validate finite element or analytical vibration models. Laser Doppler vibrometry (LDV) and high-speed digital image correlation (DIC) have become dominant methods for experimental mode shape measurement. However, vibration-based defect localization requires measurement with both high spatial resolution and accuracy. Unfortunately, the resolution of DIC is not enough to detect data anomalies caused by structural defects. On the other hand, LDV cannot achieve high-precision defect localization due to limited spatial resolution. We propose a defects localization method based on LDV-enhanced full-field vibration signatures obtained by normal-rate cameras. The experiment proves that the spectrum of LDV offers a guideline to remove the noise of cameras and can dramatically improve the vibration measurement resolution. The data fusion results show that reference-free defect localization on various structures can be easily localized.

Invite Speaker I

Mr. Max Liebmann

HOLOEYE Photonics AG

Title: LCOS SLMs as Versatile Tools for Optical Imaging and Metrology

Time: 14:15-14:45, Mar. 12 | **Onsite Room:** Changi Room 2 | **ZOOM ID:** 82976000510

Biography: Max Liebmann received the B.E. degree in physical science from the college of technology Wildau (Germany) in 2017.
He pursued with a M.E. degree in photonics from the colleges of technology Wildau and Brandenburg (Germany) in 2020.
During his study he worked as a Research Assistant at the Max-Born-Institute for Nonlinear Optics and Short Pulse Spectroscopy (Berlin, Germany). His investigations on generation and spectral characterization of optical vortices drove his interest into spatial light modulation.
Since 2020, he has been with HOLOEYE Photonics AG (Berlin, Germany) where he currently holds the position as an application manager. From there he was able to extend his research on applications for liquid-crystal on silicon display technology.

Abstract: Over the past decades, Spatial Light Modulators (SLM) have proven themselves to be a powerful device in many applications like microscopy, telecommunication, digital holography and more.
Among SLM technologies, Liquid Crystal on Silicon (LCoS) enables dynamic shaping of light fields in terms of amplitude, phase, or polarization. Being able to individually control these parameters, opens for new ideas on possible use cases. The broad field of machine vision systems is one of them.
In fact, considering already established practices, LCoS SLMs can offer solutions and improvements for many aspects of such systems: First, dynamic, and structured illumination. Second, imaging through scattered media, and ghost imaging. Third, optimization of data processing speed via optical computing.
In this regard, utilizing the versatility of LCoS SLMs, one might just overcome most of the challenges of machine vision.

Invite Speaker II

Prof. Yoshio Hayasaki

Utsunomiya University

Title: Holographic Femtosecond Laser Processing Enhanced with Optical Interferometry

Time: 14:45-15:15, Mar. 12 | **Onsite Room:** Changi Room 2 | **ZOOM ID:** 82976000510

Biography: Yoshio Hayasaki received his PhD degree in Applied Physics from The University of Tsukuba, Japan in 1993. From 1993, Prof. Yoshio Hayasaki was invited to be the faculty of Photodynamic Research Center, RIKEN, Japan, The University of Tokushima, Japan, and Utsunomiya University, Japan. Prof. Hayasaki's research areas are Applied optics/Quantum optical engineering / Applied optics/Quantum optical engineering / Science and Engineering / Inorganic industrial materials.

Abstract: A shaped beam is effective for increasing the processing throughput in laser grooving of silicon wafers, laser scribing of indium tin oxide films, and thin-film solar, and laser cleaving of glass substrates, because an ordinary Gaussian beam gives undesired melting of the material due to thermal conduction, called the heat-affected zone, occurs around the processed area if the intensity at both ends of the Gaussian profile is below the ablation threshold of the substrate material. Also, the center peak intensity of the Gaussian beam may cause serious damage to the substrate. To obtain the desired depth of the processed structure in the above material laser processing, optimal processing parameters should be explored according to the target material. A complex set of parameters is usually explored experimentally. The post-processing with scanning electron microscopy and atomic force microscopy is used to optimize the set of parameters based on a high-resolution image of the processed structure. The post-processing is iteratively performed until the desired structure is obtained. Accordingly, significant time and production costs are required. To overcome these issues, in-process monitoring of the depth of the processed structure in laser drilling of silicon with picosecond pulses has been demonstrated. In our research, optical interferometric methods based on a hand-made swept-source optical coherence tomography (SS-OCT) system and a commercially available white-light interferometry (WLI) are employed. The advantage of SS-OCT is the ability to perform non-mechanical depth scanning (A-scan). Accordingly, it is possible to perform laser processing while simultaneously measuring the depth of the processed structure. The advantage of WLI is no lateral beam scanning although it needs the axial beam scanning. They can be used depending on the purposes and the target structures. The in-process monitoring of processed structures is applied to holographic femtosecond laser processing based on a computer-generated hologram (CGH) displayed on a liquid-crystal spatial light modulator (LC-SLM).

Invite Speaker III

Prof. Chau-Jern Cheng

National Taiwan Normal University

Title: Deep Learning-enabled Holographic Tomography

Time: 16:00-16:30, Mar. 12 | **Onsite Room:** Changi Room 2 | **ZOOM ID:** 82976000510

Biography: Chau-Jern Cheng received his PhD degree in Electro-Optical Engineering from the National Chiao-Tung University, Taiwan, in 1994. From 1994 to 2007, Prof. Chau-Jern Cheng was invited to be the faculty of St. John's & St. Mary's University, Feng-Chia University, and National Taipei University of Technology. Prof. Cheng is currently the Distinguished Professor of Institute of Electro-Optical Engineering, National Taiwan Normal University. Prof. Cheng served as the Topical/Guest Editor of Applied Optics (OPTICA), Chinese Optics Letters (CLP & OPTICA) etc.. His research interests include information optics & photonics, optical metrology, digital holography, computer-generated holography, 3-D imaging & display, artificial intelligence (AI)/deep learning (DL) and related fields.

Abstract: This talk will describes a systematic approach of deep learning-enabled holographic tomography for biological cells processing, analysis, and diagnosis through three-dimensional (3-D) cell refractive index (RI) model. The deep learning-assisted technique is applied to execute the system calibration of the holographic tomography before cell processing, analysis, and diagnosis. Deep learning also perform effective segmentation of 3-D RI cell morphology for the different cellular states under normal, autophagy, and apoptosis. For heuristic example, we show a label-free possibility of 3-D cell RI model for neuroblastoma cell death identification. The biophysical parameters of 3-D RI cell morphology are analyzed and selected for learning-based classification to identify cell death pathways. The results show that the proposed approach achieve high hit rate in identifying cell morphology and cell death states through optimized biophysical parameters.

Invite Speaker IV

Prof. Wei Wang

Heriot-Watt University

Title: Towards Polarization Sensitive Machine Vision: Stokes Correlations for Diagnostics of Random Polarization

Time: 16:30-17:00, Mar. 12 | **Onsite Room:** Changi Room 2 | **ZOOM ID:** 82976000510

Biography: Wei Wang received his BSc and DSc in Physics from University of Science and Technology of China, in 1997 and 2001, respectively, and his PhD with summa cum laude in Optics from the University of Electro-Communications, Japan, in 2006. He was a Japanese Government (MONBUKAGAKUSHO) Scholar from 2002 to 2003, Research Fellow of Japan Society for the Promotion of Science (JSPS) from 2003 to 2006, and COE (Center of Excellence) Post-Doctoral Research Fellow from April of 2006. Recently, he took up a lectureship in Department of Mechanical Engineering at Heriot-Watt University, Edinburgh, UK. Dr. Wang's research interests are around statistical optics, optical information processing, imaging science and their applications to optical metrology and sensing. He has published over 60 papers in journals and conference proceedings, and was awarded the best science paper award of the Japan Society for Applied Physics in March 2007 (its highest research award of Optical Society of Japan). He is a member of the SPIE, OSA, and JSAP.

Abstract: In system engineering, machine vision has been used to extract information from an image on an automated basis for applications such as automatic inspection and robot and process guidance in industry, for security monitoring and vehicle guidance. It has been recognized that the polarization and coherence properties of light have significant influence on the image formation. With the recent progress in device technology for highresolution polarization cameras and liquid-crystal-based spatial polarization modulators, the field of machine vision and application has been growing at a fast pace and an increasing interest has been attracted for new applications such as biomedical diagnosis, remote sensing, and mineralogy.

In this talk, we will review our recent work on the Stokes-correlation-based technology with its application to machine vision. With the aid of the ensemble-average van Cittert-Zernike theorem for the propagation of polarcoherence, we investigated the autocorrelation functions and power spectra of the Stokes parameters to highlight the dependence of the polarization-related scale-size distributions. A generalized concept of the Stokes ensembleaverage coherence areas has been introduced to deal with the polarization-related average areas associated with polarization speckle. Noting the fact that the Stokes parameters cannot be measured at an ideal point, we also made investigation of the means and variances of the integrated Stokes parameters in polarization speckle after introduction of four parameters, i.e., the numbers of degrees of freedom for the Stokes detection. Furthermore, a new scheme of interferometry referred to as the Stokes Vector Interferometry has been proposed as an extension of intensity interferometry by taking the vector nature of the electromagnetic field into account. Through the cross-covariance computations for the fluctuating Stokes parameters, the squared moduli of the generalized Stokes parameters are reconstructed for the first time.

Invite Speaker V

Prof. Xiang Peng

Shenzhen University

Title: Making Range and Shape Scanning More Smart

Time: 17:00-17:30, Mar. 12 | **Onsite Room:** Changi Room 2 | **ZOOM ID:** 82976000510

Biography: Xiang Peng received his B.S., M.S., and Ph.D. from Tianjin University in 1981, 1984, and 1989, respectively, all in Optical Engineering. From July 1985 to July 1986, he is a visiting scholar at the Department of Electrical and Electronic Engineering, University of Huston, TX, USA. From August 1990 to June 1992, Dr. Peng was awarded a fellowship by Alexander von Humboldt Foundation, Germany. He worked with Institute of Applied Optics (ITO) at University of Stuttgart as an Alexander von Humboldt Fellow. Prof. Peng's current research interests include Optical Imaging, Metrology, and Optical Security. Prof. Peng has numerous publications in refereed journals and was awarded more than 20 inventions wherein ten of patents have been transferred to ESUN Co. Ltd.. He also serves as Director of Engineering Laboratory for 3D imaging and modeling of Shenzhen Government and Principal Scientist at the ESUN Co. Ltd. Prof. Peng's research activities have been supported by Natural Science Foundation of China (NSFC), Sino-German Center for Research Promotion (SGCRP), and Industrial Sectors.

Abstract: In this talk we focus on a topic of 3D scanning of complex shapes with a mobile device. When scanning an object with a handheld 3D scanner, we must deal with two key issues: sensor tracking and re-localization in order to provide a good initial guess, for the registration of multi-view range images, with use of the ICP strategy. We shall discuss the technical route of sensor tracking based on an unsupervised learning approach, following by a route of re-localization with help of fast point feature histogram (FPFH). Furthermore, we propose a learning-based approach to construct feature descriptor on the set of point clouds, which shows a promising potential for making the handheld 3D scanning more smart.

Invite Speaker VI

Prof. Rong Su

Shanghai Institute of Optics and Fine Mechanics, Chinese Academy of Science

Title: Fourier optics models for 3D image formation in interference microscopy

Time: 17:30-18:00, Mar. 12 | **Onsite Room:** Changi Room 2 | **ZOOM ID:** 82976000510

Biography: Prof. Rong Su is currently with Shanghai Institute of Optics and Fine Mechanics, Chinese Academy of Sciences. He received BSc from Sun Yat-sen University, and MSc in photonics and PhD in optical metrology from KTH Royal Institute of Technology in Sweden. Between 2015 and 2021 he worked at the National Physical Laboratory and University of Nottingham in the UK. He has had a long-term commitment to ultra-precision optical surface topography measuring instruments and technology, focusing on the basic theory of interferometric imaging and scattering, algorithm, calibration, industrial applications and corresponding international standards. He has published 40+ peer-reviewed journal papers and 2 book chapters, and had ten-year research and management experience in EU and UK projects. His research outcomes have been adopted by world-leading optical instrument manufacturers. In 2021, he was awarded a fellowship from the Chinese Academy of Sciences and since then has secured more than 20-million funding for research. He is an editor of Light: Advanced Manufacturing and Nanomanufacturing and Metrology, a program committee member of SPIE, ASPE and EOSAM conferences, an observer of the National Professional Standardization Committee, a member of the Professional Committee of Measuring Instruments of the Chinese Society of Measurement.

Abstract: Interference microscopy is a three-dimensional (3D) imaging technique that provides quantitative analysis of industrial and biomedical specimens. Although the major components of the 3D imaging theory for interference microscopy have been developed and documented, the use of the 3D imaging theory in the modelling, design, development and improvement of 3D interferometric imaging methods is advancing slowly, mainly due to the difficulty of understanding the 3D theory, inconsistencies in the previous derivations, and the missing links between scattering and imaging processes, between the 2D and 3D imaging theories, and among different approximations in the context of interference microscopy. Advance in understanding the 3D image formation in interference microscopy is introduced in this talk, providing a significant physical insight into the instrument response of objects with complex geometries, underpins the development of virtual optical instrument for surface metrology, and paves the way for new instrument designs.

Invite Speaker VII

Prof. Qian Kemao

Nanyang Technological University

Title: Data Acquisition and Analysis for Intelligent Optical Measurement

Time: 18:00-18:30, Mar. 12 | **Onsite Room:** Changi Room 2 | **ZOOM ID:** 82976000510

Biography: Dr Qian Kemao is an Associate Professor in the School of Computer Science and Engineering (SCSE) at Nanyang Technological University (NTU). He graduated from University of Science and Technology of China (USTC), where he got his BE, ME and PhD degrees. His research interests include optical metrology, image processing, parallel computing, and computer vision.

Abstract: Optical measurement is extremely important in precision engineering and other areas such as metaverse. In this talk, our first focus is the fringe pattern profilometry (FPP) for data acquisition, especially the systematic and theoretical analysis which leads to the improvement of the measurement precision. Afterwards, performance of different FPP systems will be compared. Our second focus is windowed Fourier transform (WFT) and its machine learning rivals for data analysis, to reveal their respective performances and to discuss their potentials. These efforts are expected to contribute to intelligent optical measurement for industry.

Invite Speaker VIII

Prof. Wenqi He

Shenzhen University

Title: Recent works on the GROUND GLASS: From Encrypting to Imaging

Time: 9:40-10:10, Mar. 13 | **Onsite Room:** Changi Room 2 | **ZOOM ID:** 82976000510

Biography: Associate Prof. Dr. HE works at College of Physics and Optoelectronic Engineering of Shenzhen University, China. He received his BS in optoelectronic information engineering from South China Normal University in 2007, an MS in physical electronics, and a Ph.D. in optical engineering from Shenzhen University in 2010 and 2012. Dr. He's research interests include computational optical imaging, metrology, and machine vision. He has hosted/participated a series of research projects including three China-German joint projects, and authored more than 50 papers in peer-reviewed international journals.

Abstract: Almost three decades ago, the ground glass was used to encrypt 2D information by adopting a conventional 4f architecture, namely the Double Random Phase Encoding, the milestone work in optical encryption. In the past ten years, we published a series of works to analysis the security flaws of those optical encryption schemes. About 5 years ago, we realized that the cryptanalysis is the same thing with the IMAGING through a ground glass. From then, we are mostly focusing on the imaging things, mainly using the speckle auto-correlation algorithm, the PSF-based technique or even the deep learning strategy. A crucial issue there is the limited FOV. We've made some progresses on it. In a recent work we even realized the customization of the FOV in a scattering imaging system. More recently, we find back the security/encryption application referring to the ground glass, a high-security level storage method for traditional digital secret key.

Invite Speaker IX

Prof. Percival Almoro

University of the Philippines Diliman

Title: Multiple-plane Phase Retrieval using Nature-Inspired Algorithm

Time: 10:10-10:40, Mar. 13 | **Onsite Room:** Changi Room 2 | **ZOOM ID:** 82976000510

Biography: Dr. Percival Flavier Almoro is a professor at the National Institute of Physics, University of the Philippines Diliman. He researched on holograms, phase retrieval, and optical metrology during his postdoc fellowship stints at the University of Utsunomiya, Japan in 2014, Risoe National Laboratory, Denmark in 2008-2009, and at the University of Stuttgart, Germany in 2006.
A current focus of his research is on the development of novel algorithms for enhanced phase retrieval. Stemming from object-related and detector sampling issues, lack of intensity variation in the measurements may lead to the problems of slow convergence and poor reconstruction. Assessment of these physical issues is critical and further development of more robust algorithmic strategies is being actively pursued.

Abstract: In conventional iterative phase retrieval with an initial guess phase, the lack of intensity variation can result in stagnation artifacts. Here, an algorithm based on insect behavior that naturally determines optimum solution is explored for multiple-plane phase retrieval. The effectiveness of the technique is investigated in terms of rate and quality of reconstructions.

Invite Speaker X

Prof. Peng Gao

Xidian Univeristy

Title: Quantitative phase contrast tomography (QPCT)

Time: 11:00-11:30, Mar. 13 | **Onsite Room:** Changi Room 2 | **ZOOM ID:** 82976000510

Biography: Prof. Dr. Peng Gao, studied Physics and received his Ph.D. at the Xi'an Institute of Optics and Precision Mechanics (XIOPM), CAS, in 2011. He was a "Humboldt Fellow" in University Stuttgart (2012-2014) and Marie-Curie Fellow (IEF) in KIT (2014-2018). He is currently a PI at Xidian University. His group focuses on developing quantitative phase microscopy and super-resolution optical microscopy techniques for biology. So far, he has authored over 100 peer-reviewed papers published in journals, including Nat. Photonics, Adv. Opt. Photon. Some of his publications were highlighted by tens of international media, such as Science Daily, Physics News, and so on. He is currently one of the associate editors of Optics and Laser Technology (OLT) and Frontiers in Physics.

Abstract: Optical three-dimensional (3D) microscopy is one of the indispensable common basic technologies in many frontier fields and has essential scientific and practical values. In this talk, quantitative phase contrast tomography (QPCT) for observing 3D subcellular organelles inside transparent samples (for instance biological cells) will be presented. QPCT was performed by switching LEDs distributed on a ring to generate oblique illuminations at different angles and carrying out quantitative phase contrast imaging for each illumination with a ultra-fast spatial light modulator (SLM). A 3D tomographic phase map of transparent samples can be reconstructed from the phase-shifted phase contrast images obtained at different illumination angles. We demonstrated QPCT with 3D imaging of 200-nm polystyrene microspheres (PMs) and sub-organelles inside COS7 cells. The results reveal that QPCT has a high spatiotemporal resolution, low speckle noise, and thence, it has a great potential to be applied to industrial testing and life science research.

Invite Speaker XI

Prof. Caojin Yuan

Nanjing Normal University

Title: The Light Field Modulation Techniques and Their Applications in Information Security

Time: 11:30-12:00, Mar. 13 | **Onsite Room:** Changi Room 2 | **ZOOM ID:** 82976000510

Biography: Caojin Yuan received her Ph.D in Optical engineering from Nankai University in 2008. She is a professor in the school of Physics and Technology at Nanjing Normal University. She serves as the dean of Optoelectronic information Science and Engineering Department. Her current research interests include: quantitative phase imaging and optical information encryption.

Abstract: Orbital angular momentum (OAM) holography which has attracted wide attentions from researchers in high-security encryption field. In this presentation, besides reviewing the traditional OAM holography, newly developed OAM holography techniques and their applications will be introduced. The modulated parameters including the moiré phase plate, the topological charge (TC) number, the moiré constant, the azimuthal displacement factor, the sub-region space location and the displacement amount on the structure of the moiré phase plate can be used as encoding and decoding keys in OAM holography. The feasibility of the proposed schemes with the above parameters are verified by numerical simulations and experiments. The advantage is of higher design freedom and security compared to the existed previously proposed orbital angular momentum multiplexed holography techniques. The flexibility of the moiré phase plate promotes the encrypted channel diversity and shows a new perspective.

Invite Speaker XII

Prof. Jun Ma

Nanjing University of Science and Technology

Title: Interferometric Absolute Testing Method Based on Tilted Wave Illuminations

Time: 12:00-12:30, Mar. 13 | **Onsite Room:** Changi Room 2 | **ZOOM ID:** 82976000510

Biography: Dr. Jun Ma is a professor in the school of Electronic and Optical Engineering at Nanjing University of Science and Technology. He received his B.E. in 2005, and Ph.D. in 2011 at the same university. His research interests include optical interferometry, digital holography, and imaging optics. Currently, he is a Changjiang Junior Scholar awarded by the Ministry of Education of the People's republic of China. Since 2022, he serves as the deputy dean of the school.

Abstract: The high-precision surface of the optical elements is an important guarantee for the overall performance of the optical system, and its implementation requires multiple iterations between fabrication and testing. Fizeau interferometry is commonly utilized in optical workshops as a testing technique, but the reference surface and system errors limit the ability to improve accuracy further. Therefore, interferometric absolute testing methods are required to eliminate these effects. Traditional absolute testing methods suffer from adjustment-induced errors such as translation and rotation operations to the flat, which will break the stability and introduce additional measurement errors, especially for large aperture interferometry. Different from those, this study focuses on the modulation of light source rather than the cavity and provides more conditions based on the manipulation of the light source. Specifically, a unique light source array is designed to generate four tilted wave illuminations, which can realize the micro-translation between the two coherent wavefronts. Besides, simultaneous collection of four interferograms in a single frame is realized with a novel design for the imaging optics. Combined with the traditional conjugate differential absolute testing wavefront reconstruction algorithm, high-accuracy and repeatability absolute surface results can be obtained. The proposed method avoids adjustments to the cavity and provides a novel approach for interferometric absolute testing.

Session 1 (Onsite)

Visual Based Object Recognition and Detection Technology 14:00-16:00, Mar. 13 Chair: Percival Almoro, University of the Philippines Diliman	
14:00-14:15 **M013-A**	**Application of Stereovision System for Aviation Obstacles Detection** **Michał Jurkiewicz** AP-Tech
14:15-14:30 **M027**	**Predicting Stenosis Severity and Localization in Coronary Artery using Deep Learning** **Masaki Aono**, Tetsuya Asakawa, Hiroki Shinoda, Kazuki Shimizu, Takuya Togawa, Kei Nomura Toyohashi University of Technology
14:30-14:45 **M028**	**View Stabilized Skeleton Joint Descriptor and ResNet for HAR from Skeleton Joints** **L. Nirmala Devi**, K.V.Subba Reddy, A .Nageswar Rao Osmania University
14:45-15:00 **M029**	**Spatiotemporal Modeling Encounters 3D Medical Image Analysis: Slice-Shift UNet with Multi-View Fusion** **Cynthia Ifeyinwa Ugwu**, Sofia Casarin, Oswald Lanz Free University of Bozen-Bolzno
15:00-15:15 **M033**	**Enhancing Direct Visual Odometry with Deblurring and Saliency Maps** **Magnus Kaufmann Gjerde**, Kamal Nasrollahi, Thomas Moeslund, Joakim B. Haurum Aalborg University
15:15-15:30 **M048-A**	**Why Not Normalise the Distribution for Multi-view Stereo Refinement** Ziyang Chen, **Wenting Li**, Yongjun Zhang Guizhou University; Nanyang Technological University
15:30-15:45 **M051**	**Does Hard-Negative Contrastive Learning Improve Facial Emotion Recognition?** **Khin Cho Win**, Zahid Akhtar, C. Krishna Mohan Indian Institute of Technology Hyderabad
15:45-16:00 **M067**	**Improving Optical Character Recognition in License Plate Images using Reinforcement Learning** **Aakash Gnanavelu**, Ching Yee Huang, Tze Swen Yee, **Hongyi Wang**, Oliver Aik Dee Oh, Kai Rong Chang, Siaw-Lang Wong HELP International School

Session 2 (Onsite)

Image Based Data Analysis and Application System 16:30-18:15, Mar. 13 Chair:	
16:30-16:45 **M007**	**Style-Driven Image Enhancement for Entry-Level Mobile Devices** **Neil Patrick Del Gallego**, Angelo Christian Matias De La Salle University
16:45-17:00 **M008**	**A Unified Database Schema for Geometric and Semantic Data: Continuous Volumetric Stocktaking in Gravel Quarries** **Marco Wallner**, Matthias Schörghuber, Markus Hofstätter AIT Austrian Institute of Technology
17:00-17:15 **M016**	**Selection of Image Classifiers for Noisy Images Through Metalearning** **Jens de Hoog**, Ali Anwar, Peter Hellinckx, Siegfried Mercelis University of Antwerp
17:15-17:30 **M035**	**DDS-NAS-Bench: Towards Predictors under Data Distribution Shift** **Sofia Casarin**, Cynthia Ifeyinwa Ugwu Free University of Bozen-Bolzano
17:30-17:45 **M052-A**	**Spatio-temporal Segmentation of Repeated Distinct Motions in Video Sequences with Efficient Orientation Estimation Using DgCHLAC** **Fumito Yoshikawa** Biwako Seikei Sport College
17:45-18:00 **M066**	**Temporal Change Studies in Kaziranga National Park** **Prashanth Janardhan**, Harish Narayana, Vindhya Srinivas National Institute of Technology Silchar
18:00-18:15 **M069**	**Diffusion and Multi-Domain Adaptation Methods for Eosinophil Segmentation** **Kevin Lin**, Donald Brown, Sana Syed, Adam Greene University of Virginia

Session 3 (Online)

Intelligent Image Analysis and Methods 10:30-12:00, Mar. 14 | ZOOM ID: 82976000510 Chair: Le Hong Trang, Ho Chi Minh City University of Technology	
10:30-10:45 **M030-A**	**A Comparative Study between AI-based and Manual Inspection Techniques** **Wei Siong Tan**, Blaine Bateman, Hsiao-Kuang Yu Glow-Ai Inc
10:45-11:00 **M017**	**Prompt Retrieval from Stable Diffusion Generated Images Using the ConvNeXt CLIP Network** **Weijing Cheng** Nanjing Jinling High School
11:00-11:15 **M018**	**Research on Intelligent Control Methods for Community Real Population Based on Big Data and Object Detection** **Zheng Ren**, Yingqiu Li, Yujun Shen, Yanxin Xu, Dalian Neusoft University of Information
11:15-11:30 **M064**	**Performance Enhancement of Covid-19 Chest X-Ray Image Classification Using GAN and CNN** **Sanjeeb Prasad Panday**, Mani Prasad Dumre, Aman Prasad Shakya, Basanta Joshi Tribhuvan University
11:30-11:45 **M068**	**A Breast Cancer Prognosis Model using PyRadiomics and Image Segmentation from MRI Data** **Moirangthem Sonia**, Indrajit Kalita, Debashree Devi Indian Institute of Information Technology Guwahati
11:45-12:00 **M039**	**Let's Talk About Image Quality Improvement in Underwater Images for Object Detection** **Kirill Bogomasov**, Tim Geuer, Stefan Conrad Heinrich Heine University

Session 4 (Online)

Image and Data Analysis Based on Machine Learning 14:00-15:30, Mar. 14 | ZOOM ID: 82976000510 Chair: Burra Venkata Durga Kumar, Xiamen University Malaysia	
14:00-14:15 **M011**	**Hostile Vessel Behavior Prediction using LSTMs** **Ayesh Meepaganithage**, Mircea Nicolescu, Monica Nicolescu University of Nevada Reno
14:15-14:30 **M043**	**VQUNet: Vector Quantization U-Net for Defending Adversarial Attacks by Regularizing Unwanted Noise** **Zhixun He**, Mukesh Singhal University of California
14:30-14:45 **M010**	**SERF-P: A Semi Supervised Approach via Enhancing Representative Features and Pseudo-Labeling** **Tin Duong Duc**, Phuong Thai Tieu, Trang Le Hong Ho Chi Minh City University of Technology
14:45-15:00 **M041**	**Deep Learning-based Land Use and Land Cover Changes Detection from Satellite Imagery : a case study of the city of Richard Toll** **Mandicou BA**, Pape Ibrahima THIAM, Etienne DELAY, Charles Abdoulaye NGOM, Idy DIOP, Alassane BAH Cheikh Anta Diop University
15:00-15:15 **M060**	**Recovering Image Information from Colored Speckle under RGB Illumination in an Imaging System** **Shuang Liu**, Steen G. Hanson, Mitsuo Takeda, Wei Wang Xi'an Technological University
15:15-15:30 **M061**	**Modeling Technique for Optimizing Simulation Objective Function Based on Key Parameters of the X-ray Imaging System** **Zhiyu Gao**, Haibin Lan, Changsheng Zhang, Yao Huang, Jian Fu Beihang University

Posters

16:00-16:30, Mar. 13 @ Foyer	
M031	**Research on Engineering Management Question-answering System in the Communication Industry Based on Large Language Models and Knowledge Graphs** **Yingdi Jiang**, Jiarui Yao, Fangfei Li, Yan Zhang China Telecom Corporation Limited Beijing Branch
M034	**FAIC-Attack: An Adversarial Watermarking Attack against Face Age based on Identity Constraint** **Xuning Zheng**, Xiankang Wang, Ziyi Yu, Siyu Xia Southeast University
M037-A	**A Weathered Texture Generation Approach Based on Deep Learning Techniques** **Chin-Chen Chang**, Yao-Min Chen National United University